

TABLE 17 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR SAMPLE PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	2.86	3.34	3.90	4.56	5.32	6.23	7.32	8.63	17.2%
Japan	1.33	1.56	1.84	2.15	2.53	2.98	3.51	4.16	17.8%
India	0.17	0.20	0.23	0.27	0.31	0.36	0.43	0.50	16.9%
Rest of Asia Pacific	2.40	2.79	3.25	3.78	4.40	5.13	6.00	7.05	16.8%
Total	6.76	7.90	9.22	10.76	12.56	14.71	17.27	20.35	17.2%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 18 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SAMPLE PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.72	0.80	0.89	0.98	1.08	1.20	1.34	1.49	11.0%
Rest of Latin America	0.82	0.90	1.00	1.10	1.21	1.35	1.49	1.66	10.8%
Total	1.54	1.70	1.88	2.08	2.30	2.55	2.83	3.16	10.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

6.3 SEQUENCING AND LIBRARY PREPARATION

6.3.1 SEQUENCING AND LIBRARY PREPARATION SERVICES TO DOMINATE MARKET DURING FORECAST PERIOD

This segment includes sequencing and library preparation services for samples subjected to microbiome sequencing. This is a crucial step in microbiome analysis, wherein either reference sequences are created for samples or sequencing outputs are compared against the existing sequences. Different approaches are employed in microbiome sequencing and library preparation, including 16S rRNA sequencing, shotgun metagenomics, qPCR testing, and short-chain fatty acids (SCFA) analysis. Service providers use either state-of-the-art equipment and software or purchase commercial systems and software solutions. Illumina, Inc. (US) and PacBio (US) are among the popular brands adopted by a majority of service providers to offer sequencing and library preparation services. The expanding pool of service providers, particularly the ones offering NGS services, is the key contributor to the dominance of this segment.

TABLE 19 MICROBIOME SEQUENCING SERVICES MARKET FOR SEQUENCING AND LIBRARY PREPARATION, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	61.38	69.63	78.94	89.58	101.75	115.96	132.62	152.34	14.1%
Europe	33.26	38.09	43.60	49.94	57.25	65.87	76.04	88.17	15.1%
Asia Pacific	9.17	10.79	12.68	14.91	17.53	20.67	24.44	29.01	18.0%
Latin America	2.09	2.33	2.60	2.90	3.23	3.61	4.04	4.54	11.8%
Middle East and Africa	1.45	1.60	1.76	1.93	2.13	2.34	2.58	2.85	10.2%
Total	107.34	122.43	139.58	159.26	181.88	208.45	239.73	276.91	14.7%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 20 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SEQUENCING AND LIBRARY PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	55.63	63.15	71.65	81.37	92.49	105.48	120.73	138.78	14.1%
Canada	5.75	6.48	7.29	8.21	9.26	10.48	11.89	13.56	13.2%
Total	61.38	69.63	78.94	89.58	101.75	115.96	132.62	152.34	14.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 21 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR SEQUENCING AND LIBRARY PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	7.20	8.26	9.47	10.87	12.49	14.40	16.66	19.35	15.4%
UK	6.44	7.43	8.56	9.87	11.40	13.20	15.35	17.92	15.9%
France	5.09	5.83	6.67	7.65	8.77	10.10	11.66	13.53	15.2%
Italy	1.43	1.64	1.87	2.14	2.44	2.80	3.23	3.73	14.8%
Spain	1.10	1.26	1.43	1.64	1.87	2.14	2.46	2.84	14.7%
Rest of Europe	12.00	13.68	15.59	17.77	20.28	23.23	26.69	30.80	14.6%
Total	33.26	38.09	43.60	49.94	57.25	65.87	76.04	88.17	15.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 22 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR SEQUENCING AND LIBRARY PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	3.88	4.56	5.36	6.30	7.41	8.74	10.33	12.26	18.0%
Japan	1.80	2.14	2.52	2.98	3.53	4.18	4.97	5.93	18.6%
India	0.23	0.27	0.32	0.37	0.44	0.51	0.61	0.72	17.8%
Rest of Asia Pacific	3.25	3.82	4.48	5.25	6.16	7.24	8.54	10.10	17.7%
Total	9.17	10.79	12.68	14.91	17.53	20.67	24.44	29.01	18.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 23 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SEQUENCING AND LIBRARY PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.98	1.10	1.22	1.36	1.52	1.70	1.91	2.14	11.9%
Rest of Latin America	1.11	1.24	1.38	1.53	1.71	1.91	2.14	2.40	11.8%
Total	2.09	2.33	2.60	2.90	3.23	3.61	4.04	4.54	11.8%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

6.4 DATA ANALYSIS

6.4.1 DEVELOPMENT OF ADVANCED BIOINFORMATICS SOLUTIONS TO DRIVE GROWTH

This segment includes bioinformatics/data analysis services for samples subjected to microbiome sequencing. Researchers in the genomics field often outsource the data analysis part of their projects to bioinformatics service providers. NGS data analysis services are offered for data processing and quality control, assembly, mapping, annotation, and advanced analysis for microbiome research and population studies.

The rising awareness about NGS data analysis services, the recent launch of data analysis services by bioinformatics companies, and the increasing availability of data analysis services in developed countries are the major factors supporting the growth of this market.